

npm - Installation & Setup

A complete guide to prepare the environment, install, configure, validate and perform the first functional use of the tool.

Manual objective

This manual guides you through the complete installation and setup process, from preparing the environment to final validation. Follow the steps in order and use the checks throughout the guide to confirm that each configuration was completed correctly.

Official references

<https://docs.npmjs.com/downloading-and-installing-node-js-and-npm>

What it is

npm is the package manager commonly distributed with Node.js.

Official site and documentation

Use the official npm source as the primary reference:

<https://docs.npmjs.com/downloading-and-installing-node-js-and-npm>. Official documentation should be the first source for installation, supported versions, syntax changes and new features.

package.json

It records dependencies, metadata and scripts. package-lock.json helps keep reproducible versions.

QA automation

Use npm ci in pipelines when a consistent lockfile exists; scripts can standardize linting, tests, reports and environment execution.

When to use it

Use this knowledge when product risk, architecture or the delivery stage requires this type of validation. Confirm the goal, environment, required data and expected evidence before starting.

How QA applies it

Start from the behavior or risk to validate, prepare data and environment, execute the technique or tool reproducibly and capture evidence that helps the team understand the result.

Good practices

Prioritize high-impact scenarios, keep data and environments controlled, record important decisions and review the approach periodically to avoid redundant or outdated coverage.

First QA-oriented use

Pick one small, verifiable task in npm, configure only what is necessary, run a simple example and confirm the outcome. Repeat with different data or conditions before adding abstractions.

Using it in a real project

Define where npm fits, who owns configuration, how credentials and data are handled and how results are shared. Integrate it into the team workflow instead of creating an isolated activity.

Adoption good practices

Standardize versions and configuration, document important decisions, keep secrets out of code and preserve reproducible examples. Reassess whether npm still delivers value as the product evolves.

Common issues and troubleshooting

When npm fails, verify versions, dependencies, permissions, connectivity, data and environment before assuming a product defect. Preserve the original error and reduce the issue to the smallest reproducible case.